

SubseaGuard AI

Operator & Engineering Dashboard User Guide

Welcome to the SubseaGuard AI Predictive Maintenance & Integrity Management Platform for deepwater subsea flowlines and risers.

This guide provides a step-by-step walkthrough of how to navigate the interface, interpret telemetry data, triage anomalies, and execute human-in-the-loop (HITL) condition-based interventions.

High-Level Layout Overview

The platform is designed following Schlumberger DELFI, Palantir Foundry, and Baker Hughes control-room standards for high data density, readability under time pressure, and industrial auditability.

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TIER 1: EXECUTIVE BAR
(Brand | Role Switcher | Anomaly Dropdown | Live Toggle | Audio | Architecture | Audit)
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TIER 2: NAVIGATION RIBBON
(11 Specialized Engineering Modules)
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TOP KPI STRIP
(Fleet Overall Health | MTBF | CBI OPEX Savings -$3.83M | Active Critical Alerts)
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PRIMARY WORKSPACE
(Active Role Console / 5-Zone Control Room)
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FOOTER: REAL-TIME BUS TELEMETRY
(20 Hz DAS Fiber | IEEE 1588 PTP +/-8ns | SHA-256 Audit: 8f4b2a901c3e)
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Step 1: Navigating the Top Executive Control Bar (Tier 1)

Located at the very top of your screen:

- **Subsea Asset Coordinates:** Displays your current field location (Block-4, Depth: -1,850m) with live optical telemetry ping.
- **Role Switcher:** Click between 4 tailored perspectives (see sub-list below).
 - Field Technician: Touch-friendly traffic lights, actionable SOP procedures, simplified live readouts.
 - Subsea / Offshore Engineer: Hardware barriers, catenary TDZ bending stress, VIV lock-in, CP cathodic protection, and valve stroke tests.
 - Reliability Engineer: Full time-series trends, autoencoders, Bayesian UQ bounds (P10 / P50 / P90), and SHAP explainability.
 - Asset Manager: Downtime cost risk (\$/hr), OPEX cost avoidance (-\$3.83M), and regulatory audits.
- Anomaly Scenario Selector:** Dropdown menu allowing you to inject 6 physics-informed operational scenarios in real time:
 - NORMAL: Steady-state baseline.
 - HYDRATE_RISK: Deepwater subcooling below hydrate phase boundary.
 - TOUCHDOWN_FATIGUE: Current-induced 0.38 Hz VIV lock-in on Steel Catenary Riser (SCR-01).
 - EROSION_WALL_LOSS: Wellhead sand breakthrough (84.5 PPM) causing pipe wall thinning.
 - MICRO_LEAK: Negative Pressure Wave (NPW) acoustic leak detection at KP 4.35 km.
 - WAX_DEPOSITION: Subcooling below WAT (37.5C) creating paraffin wax constriction.
- **Live Play / Pause (LIVE / PAUSE):** Toggles live telemetry simulation steps (1.5-second clock tick).

- **Audio Alarm Toggle:** Enables audible subsea acoustic pinging when high-severity anomalies are detected.
- **Architecture Button:** Opens the comprehensive 5-Stage Enterprise Architecture Pipeline modal (Ingestion → Features → Bayesian UQ → HITL Inspection → Remediation & Compliance).
- **ISO 14224 Audit Export:** Generates a certified subsea equipment taxonomy and regulatory compliance report ready for PDF export or printing.

Step 2: Mastering the 5-Zone Industrial Control Room

By default, the dashboard opens in the Control Room (5 Zones) tab — a single-pane-of-glass interface:

Zone 1: System Overview Header

- Shows target asset specifications (e.g. PFL-101 Primary Multiphase Flowline, 14-inch Pipe-in-Pipe, 12.4 km).
- Displays Overall Health Index (0-100%), Active Critical Alert Count, Operating Mode (AUTONOMOUS AI-CBI), and IEEE 1588 PTP Clock Sync status (+/-8ns jitter).

Zone 2: Live Spatial Schematic Panel (KP 0.0 to KP 12.4)

Shows a 2D/3D elevation profile of the seabed route from Manifold SM-01 (KP 0.0) to PLET-01 / SCR-01 (KP 12.4).

Overlaid Sensor Nodes:

Sensor	KP Location	Function
PT-01	KP 0.0	Inlet Quartz Pressure & Temperature
TDZ-ACC	KP 2.85	Riser Touchdown Accelerometer (VIV lock-in sensor)
DAS-NPW	KP 4.35	Distributed Fiber-Optic Acoustic Sensor & Methane Sniffer
DTS-HYD	KP 6.2	Distributed Temperature Sensing Loop
WAX-DP	KP 9.0	Differential Pressure Constriction Monitor
UT-AUT	KP 12.4	Phased Array Ultrasonic Wall Thickness Gauge

Color Coding: Green = Nominal | Amber = Approaching Limit | Red = Active Breach / Anomaly.

Zone 3: Real-Time Sensor Trend Panel

- Multi-stream time-series graph tracking Inlet Pressure (bar), Fluid Temperature (C), and Flow Rate.
- Physics Reference Bands: Dotted threshold lines show hard safety barriers (e.g. Hydrate Formation Boundary at 14.5C and ASME B31.8 Minimum Wall Limit at 15.2 mm).
- Use the Time Window buttons (5m, 1h, 24h, 7d) to adjust trend resolution and identify gradual sensor drift before threshold breaches occur.

Zone 4: Diagnostic & Bayesian RUL Panel

- **Dominant Fault Mode:** Model output identifying the specific failure mechanism (e.g., Sand Erosion & H2S Sour Pitting or VIV 2nd Harmonic Modal Lock-in) with AI confidence score (e.g. 98.4%).
- **Bayesian Uncertainty Quantification (UQ):** Displays P10 (90% conservative limit), P50 (nominal median), and P90 (optimistic bound) remaining useful life in operating days.

- **SHAP Explainability Strip:** Ranked breakdown of the exact physical parameters driving the prediction (e.g., fluid velocity +28%, sand PPM +54%, subcooling ΔT +68%).
- **Data Trust Widget:** Compares the physical ROV Ultrasonic Scan Ground Truth (23.80 mm) against the AI Weibull Projection (23.90 mm), validating model accuracy (99.58%).

Zone 5: Tiered Alert Queue & HITL ROV Dispatch

- Prioritizes active alarms by tier: CRITICAL (Immediate shut-in/inspection), WARNING (Approaching limit), INFORMATIONAL (Drift detected).
- 1-Click ROV Authorization: Click "Initiate ROV Anomaly Sweep" to launch the Human-in-the-Loop (HITL) Authorization Gate modal.

Step 3: Authorizing an ROV Inspection (HITL Workflow)

When an alert appears in Zone 5:

- 1 Click the red button: "Initiate ROV Anomaly Sweep (Engineer Sign-Off Gate)".
- 2 A verification modal will open containing: Triggering Event (specific sensor threshold or acoustic anomaly); Target Asset & Location (e.g. PFL-101, KP 4.35 or SCR-01, KP 2.85); Vessel Spread & Rate (estimated mobilization cost, e.g. DSV Deep Constructor at \$115,000/day); Tooling Payload (pre-selected tooling: Phased-Array UT, CP Dip needle, Laser Scaler, Optical Sniffer).
- 3 Sign-off Requirement: Check the mandatory compliance box "I formally authorize mobilizing the ROV spread under DNV-RP-F116 guidelines", and enter/verify the authorizing engineer name.
- 4 Click "Sign & Execute ROV Dispatch". The mission transitions to APPROVED_DISPATCHED and logs an entry in the audit trail.

Step 4: Closing the Maintenance Loop (Model Retraining)

When an offshore crew completes an inspection or intervention:

- 1 Navigate to the Alerts & Escalation tab (or click "Sign Off & Log Completed Work Order" in the Technician view).
- 2 Click "Log Completed Intervention".
- 3 Select the target asset, enter the action taken (e.g. ROV Ultrasonic Phased Array Scan Completed), and input the newly verified physical wall thickness (e.g. 24.2 mm).
- 4 Click "Submit & Retrain Model Baseline".

What happens under the hood: The platform updates the asset baseline, resets the anomaly scenario, clears active critical alerts, and lowers the Population Stability Index (PSI) drift metric back to baseline (0.012).

Step 5: Exploring Specialized Engineering Tabs (Tier 2)

Tab	Purpose & Features
Control Room (5 Zones)	Integrated single-pane control room for 24/7 subsea monitoring.
Flowline Spatial Profile	Continuous condition monitoring with interactive KP elevation profiles and multi-channel telemetry streams.

Tab	Purpose & Features
3D Seabed Digital Twin	2.5D bathymetry canvas at -1,850m with animated multiphase flow streams, catenary curves, and Pipe-in-Pipe cross-sectional FEM stress meshes.
ROV / AUV Operations	Simulated 4K subsea camera HUD with marine snow sediment, 360-degree rotating sonar radar, depth gauges, tether tension, and payload tool selection.
Fault Classification & XAI	Deep learning multi-fault confidence matrix, severity triage ranking, and zone-by-zone RUL forecasts.
Flow Assurance & Hydrate	Thermodynamic P-T hydrate phase envelope with dynamic compositional shifting (Water Cut & GOR) and bounded RL MEG chemical optimizer.
Corrosion & Wall RUL	Ultrasonic wall degradation vs ASME B31.8 limits, decomposing de Waard CO ₂ , NACE H ₂ S sour pitting, and SRB bio-corrosion (MIC).
Touchdown Fatigue & VIV	Dynamic catenary bending stress, 0.38 Hz vortex lock-in FFT spectra, Rainflow cyclic damage, and Paris crack propagation (da/dN).
Early Leak Detection	Mass balance discrepancy monitor (Q _{in} - Q _{out}), Negative Pressure Wave (NPW) acoustic time-of-flight KP pinpointing, and optical sniffer plume mapping.
Alerts & Escalation	Full historical alert logs, Twilio/SMTP SMS & email automated paging gateway, and closed-loop maintenance records.
Cost & CBI Economics	Condition-Based Inspection (CBI) ROI analytics detailing \$3.83M annual cost avoidance, 28 vessel days saved, and 640 tons of CO ₂ e mitigated.

Quick Demonstration Guide (Testing Scenarios)

To run a live demonstration of the system:

- 1 Trigger a Leak:** Select MICRO_LEAK in the top ANOMALY dropdown. Notice the alarm sound, the schematic flashing red at KP 4.35, the mass balance discrepancy rising to 1.85 kg/s, and the alert queue displaying the NPW acoustic containment breach.
- 2 Review Explainability:** Observe the SHAP strip highlighting DAS Acoustic NPW Energy: 88.5 dB as the dominant factor (+58%).
- 3 Dispatch ROV:** Click "Initiate ROV Anomaly Sweep", check the authorization box, and confirm dispatch.
- 4 Switch to ROV HUD:** Click the ROV / AUV Operations tab to watch the subsea camera stream and optical sniffer verify the hydrocarbon plume.
- 5 Close the Loop:** Log an intervention or switch back to NORMAL to restore steady-state operation.